

LANSDOWNE HOUSE, ON, Canada

WMO: 718460

Lat: 52.200N

Lon: 87.940W

Elev: 831

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.2	-29.5	-41.7	0.5	-33.9	-36.5	0.7	-28.6	17.5	7.7	16.1	6.2	3.2	260	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	82.6	65.5	79.3	63.9	76.2	62.7	69.2	77.6	67.0	74.7	64.9	72.5	10.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	99.4	73.5	64.1	92.3	70.9	62.0	85.8	68.8	33.9	77.9	32.1	74.9	30.4	72.7	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.6	15.5	13.9		-42.1	90.0	5.9	4.3	-46.3	93.1	-49.7	95.6	-53.0	98.1	-57.3	101.2	
				-41.1	73.1	5.3	2.2	-44.9	74.7	-48.1	76.0	-51.1	77.2	-55.0	78.8	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.2	-5.2	-1.4	12.7	29.2	45.0	57.5	62.8	60.1	51.1	37.3	20.1	3.0
	DBStd	26.28	14.03	13.21	14.63	12.03	10.11	8.48	6.39	6.64	8.34	8.36	12.53	14.78
	HDD50	8035	1712	1439	1157	627	224	26	2	4	85	405	898	1456
	HDD65	12460	2177	1859	1622	1074	624	252	119	183	423	858	1348	1921
	CDD50	1168	0	0	1	3	69	250	398	316	119	12	0	0
	CDD65	118	0	0	0	0	4	27	51	30	6	0	0	0
	CDH74	1165	0	0	0	1	78	322	453	258	52	1	0	0
	CDH80	246	0	0	0	0	15	75	101	47	8	0	0	0
Wind	WSAvg	6.7	6.2	6.1	6.4	6.8	7.1	7.0	6.5	6.7	7.2	7.3	6.7	6.2
Precipitation	PrecAvg	28.3	1.3	0.9	1.3	1.8	2.2	3.3	3.9	3.7	3.5	2.6	2.2	1.4
	PrecMax	36.7	4.2	2.3	3.1	4.1	3.7	9.2	9.2	8.5	6.9	7.4	4.2	2.6
	PrecMin	20.1	0.4	0.3	0.4	0.3	0.8	0.8	1.5	1.4	0.7	0.6	0.6	0.7
	PrecStd	4.4	0.9	0.6	0.6	1.1	0.7	1.8	1.9	1.9	1.6	1.3	0.9	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	36.5	54.6	66.7	81.6	86.8	87.9	84.5	80.2	68.1	52.2	35.8
		MCWB	32.5	33.7	45.0	50.0	59.3	66.5	69.3	68.7	65.4	58.5	46.0	32.7
	2%	DB	28.5	32.2	45.0	59.3	75.0	81.5	82.5	80.5	73.7	60.9	44.0	32.0
		MCWB	27.6	30.0	37.6	45.9	56.8	63.3	66.0	66.3	62.9	53.0	40.5	31.0
	5%	DB	23.6	26.9	39.7	53.0	69.2	77.7	78.7	76.5	69.1	55.2	39.6	28.7
		MCWB	22.7	24.8	34.4	41.5	54.1	61.5	64.6	64.9	60.1	49.8	37.2	27.8
	10%	DB	17.4	20.6	34.8	47.4	63.4	73.4	75.6	72.8	64.9	50.1	35.5	24.5
		MCWB	16.6	19.0	30.9	38.3	51.3	59.8	63.2	62.6	58.1	46.5	34.1	23.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	34.3	45.1	52.0	63.6	70.8	72.5	71.4	67.7	60.1	47.4	33.6
		MCDB	34.2	35.9	52.3	64.7	74.5	81.9	84.0	80.0	74.5	67.1	50.5	34.8
	2%	WB	27.6	30.1	38.5	46.2	58.7	67.0	69.4	68.6	64.6	54.2	40.7	30.9
		MCDB	28.5	32.2	43.6	57.8	70.7	75.9	77.4	77.0	72.0	58.6	43.1	32.0
	5%	WB	22.8	25.1	34.8	42.4	55.9	64.1	66.9	66.6	61.5	50.3	37.3	27.8
		MCDB	23.9	26.8	39.3	51.7	67.1	72.9	74.5	73.9	67.2	54.4	39.4	28.7
	10%	WB	16.6	19.2	31.3	38.9	52.6	61.4	64.8	64.2	58.5	46.5	34.1	23.6
		MCDB	17.4	20.7	34.6	46.6	62.7	70.7	72.7	70.4	63.8	49.9	35.5	24.5
Mean Daily Temperature Range	5% DB	MDBR	18.8	23.8	25.9	23.1	21.9	20.9	19.6	19.0	16.9	12.3	12.8	16.3
		MCDBR	19.0	22.4	26.9	30.9	31.2	27.2	25.1	24.6	23.4	22.0	15.5	15.7
		MCWBR	18.4	20.1	20.1	19.1	15.9	13.2	12.3	13.1	13.9	15.1	12.2	15.0
	5% WB	MCDBR	18.8	21.0	23.0	28.2	26.6	23.3	20.6	20.5	20.0	20.4	14.1	15.6
		MCWBR	18.3	19.3	17.6	18.1	15.0	13.0	11.8	12.3	13.4	15.0	11.7	14.9
Clear-Sky Solar Irradiance	taub	0.237	0.253	0.278	0.332	0.352	0.359	0.388	0.363	0.342	0.297	0.268	0.236	
	taud	2.301	2.297	2.316	2.346	2.433	2.440	2.391	2.462	2.515	2.579	2.448	2.315	
	Ebn at Noon	251	280	293	287	285	284	272	274	268	263	238	231	
	Edh at Noon	19	26	31	35	34	34	35	31	26	20	17	16	
All-Sky Solar Radiation	RadAvg	368	761	1230	1654	1660	1781	1718	1399	961	554	286	230	
	RadStd	37	49	96	119	141	201	150	122	105	72	33	36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page